

Diagnosing Identifiability in Two-Tower Models for Unbiased Learning to Rank

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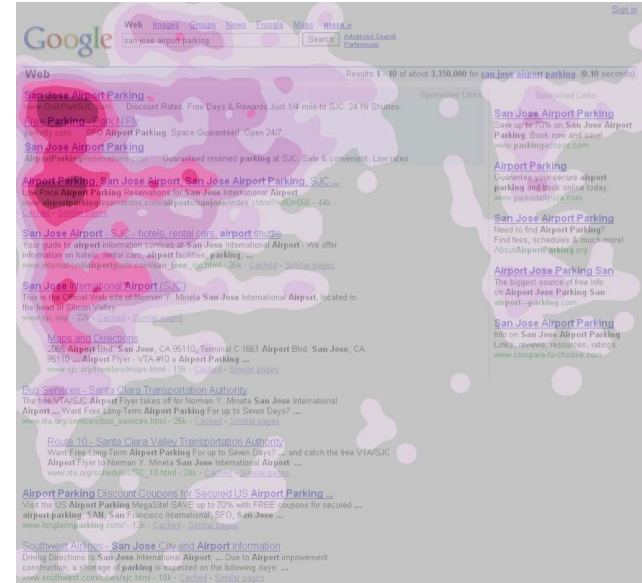


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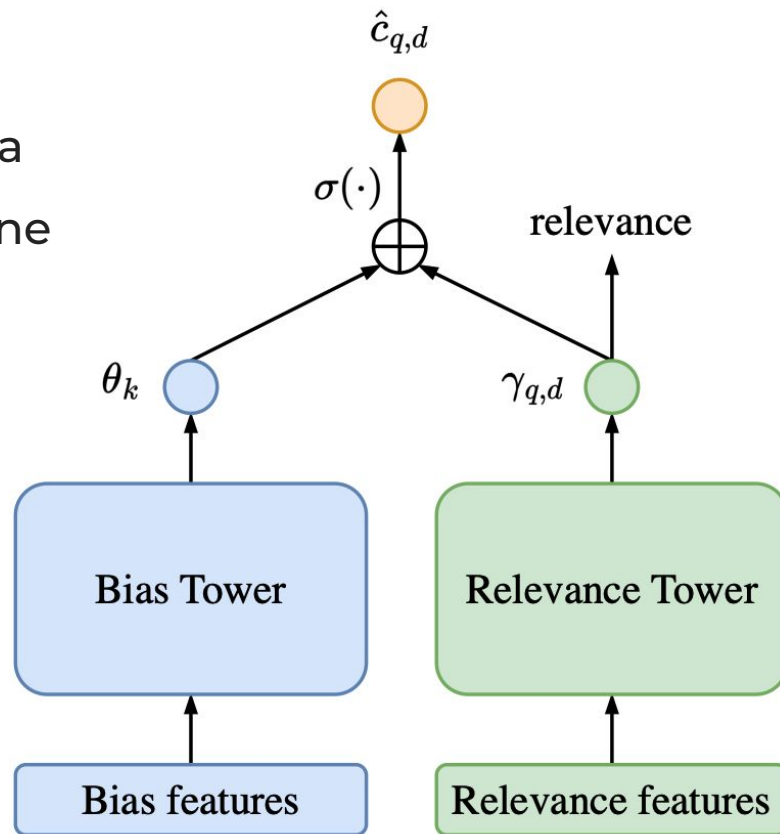
Unbiased Learning to Rank

- **Position bias:** Top-ranked items are more likely to be observed, and subsequently clicked
- A click is a **biased** relevance signal
- **Unbiased learning to rank** learns ranking models from biased clicks



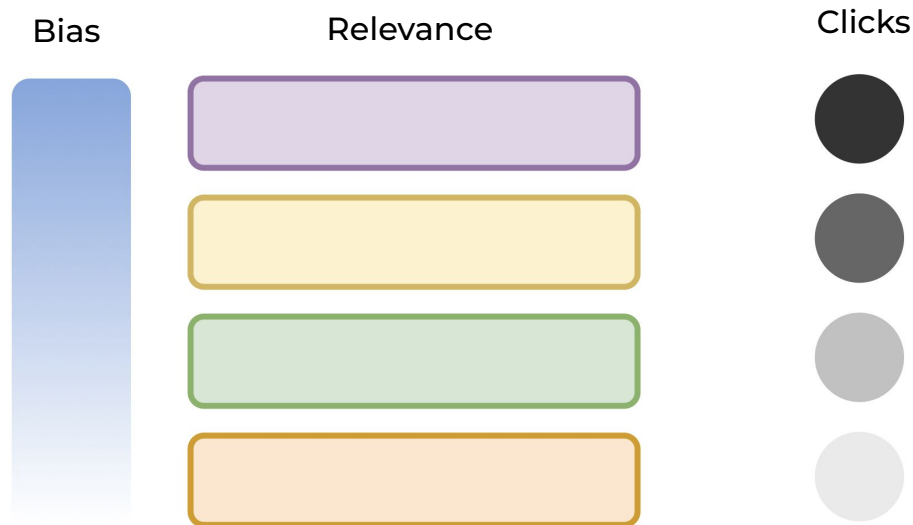
Two-Tower models

- **Learn** the probability of a click from a combination of networks (towers), one for **bias**, one for **relevance**.
- **Jointly** optimize networks on click prediction task
- Use **relevance** tower for relevance prediction in ranking tasks.



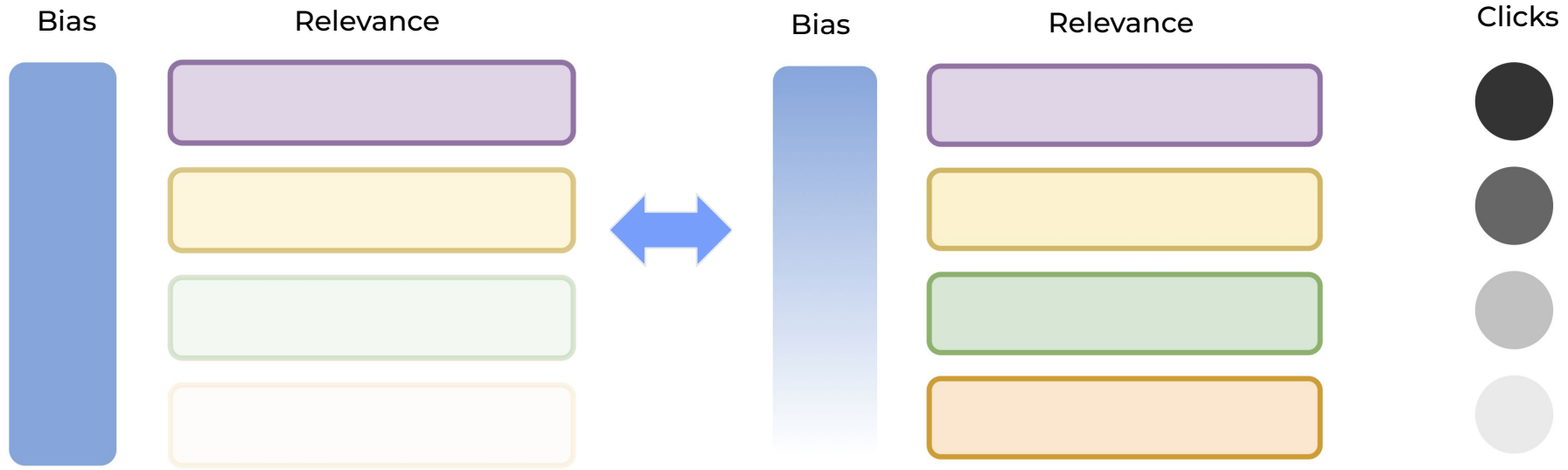
Identifiability in Two-tower models

Identifiability: Is there a **single** combination of bias and relevance parameters which explains the observed clicks?



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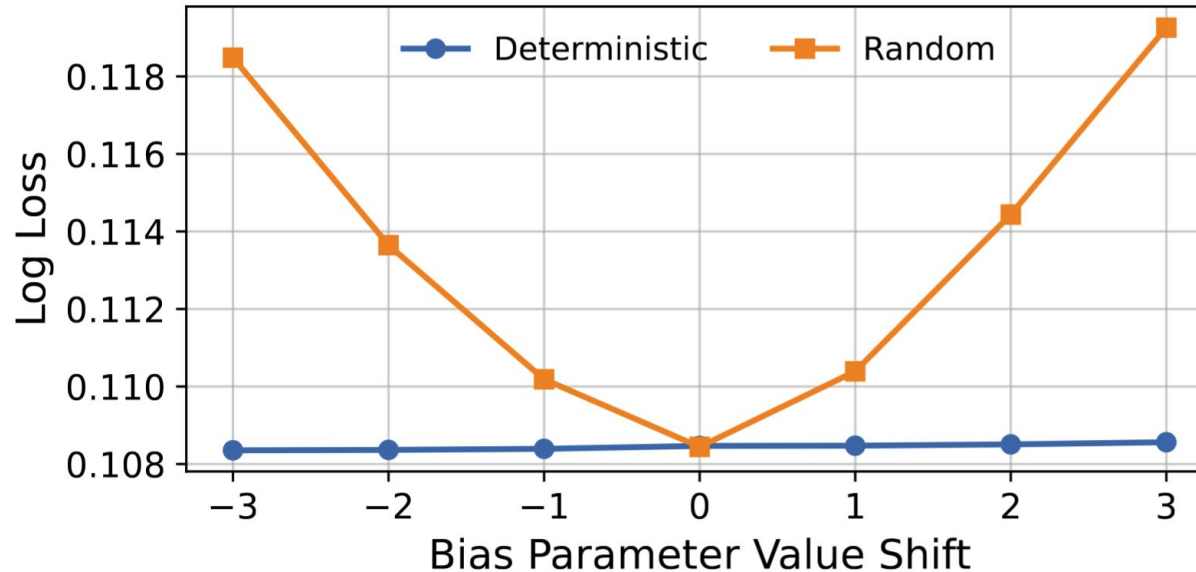
Measuring Identifiability in Two-tower models

- When is our model identifiable?
 - If we have ranking **swaps** across positions [3]
 - If we have a sufficient amount of **feature overlap** [2]
- Existing guarantees are only theoretical:
 - How many ranking swaps are sufficient for a model?
 - What is the necessary amount of feature overlap?
- How do we measure whether a **model-dataset combination** is identifiable?

[2] Hager, P., Zoeter, O., & de Rijke, M. Unidentified and Confounded? Understanding two-tower models for unbiased learning to rank. In ICTIR 2026.

[3] Mouxiang Chen et al. Identifiability matters: revealing the hidden recoverable condition in unbiased learning to rank. In ICML 2024.

The Profile Likelihood for diagnosing Identifiability



Shift and freeze the value of a bias parameter, measuring loss difference [4]

- Flat loss → Multiple Parameter Combinations → Unidentifiable
- Loss decreases with shift → Single set of optimal parameters → Identifiable

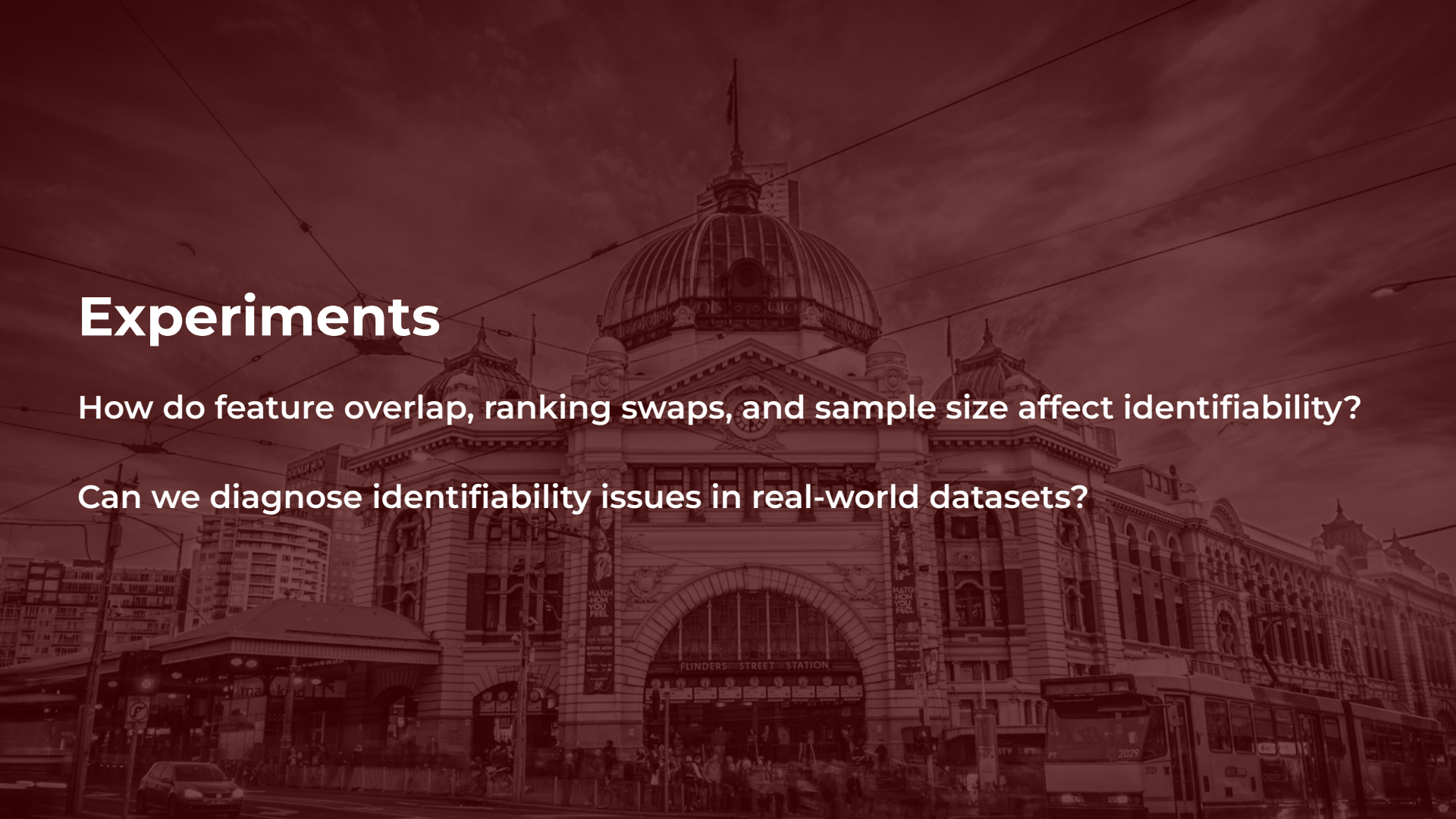
Method: Profile Likelihood for Diagnosing Identifiability

1. We shift a bias parameter away from its optimum, freeze it, and retrain the rest of the model.
2. We compare the loss of the shifted model with the loss of the original parameters.
3. Evaluate if the loss difference is statistically significant with a likelihood test
 - a. If threshold is never exceeded, we have structural unidentifiability
 - b. If threshold is exceeded on one side, practical unidentifiability
 - c. If threshold is exceeded on both sides: Identifiability is found
4. Repeat for all bias parameters to determine model identifiability

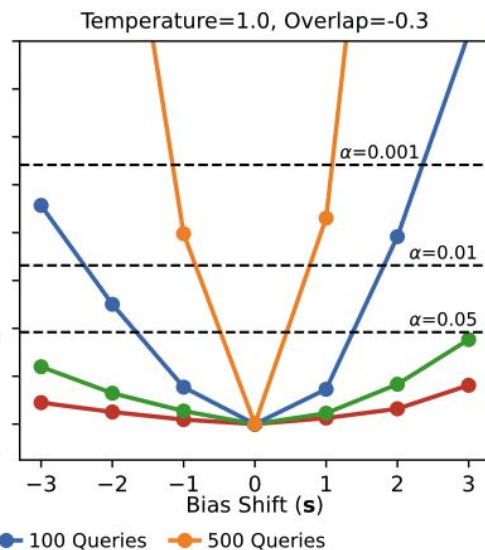
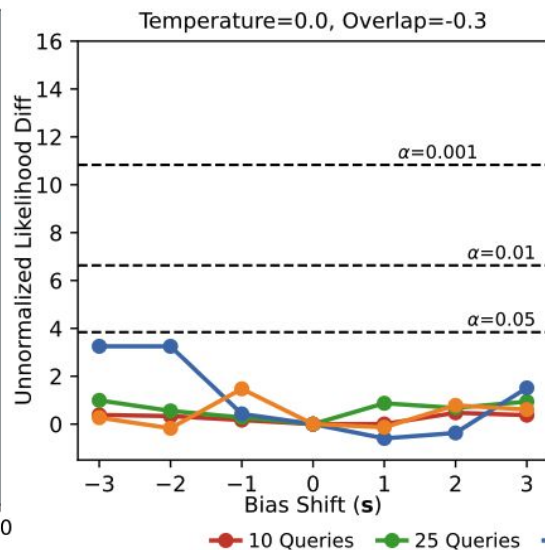
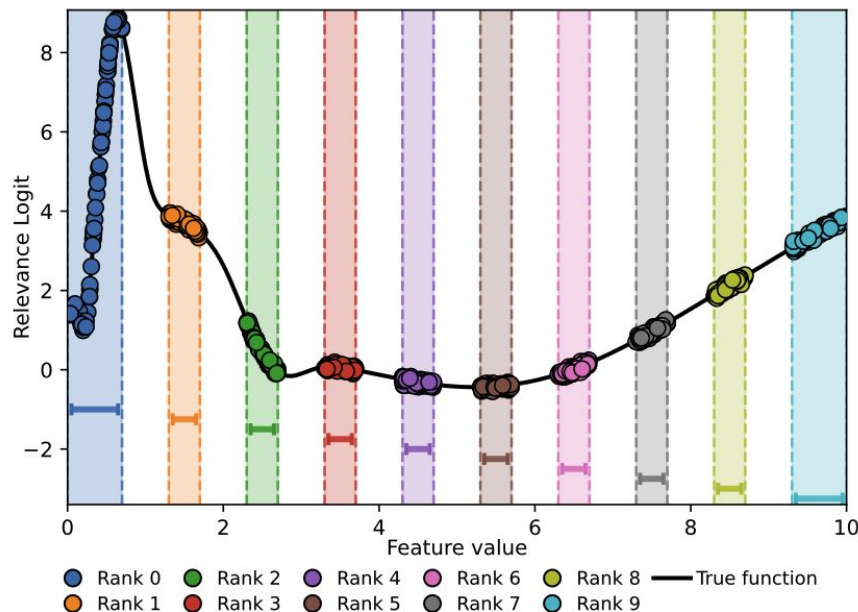
Experiments

How do feature overlap, ranking swaps, and sample size affect identifiability?

Can we diagnose identifiability issues in real-world datasets?

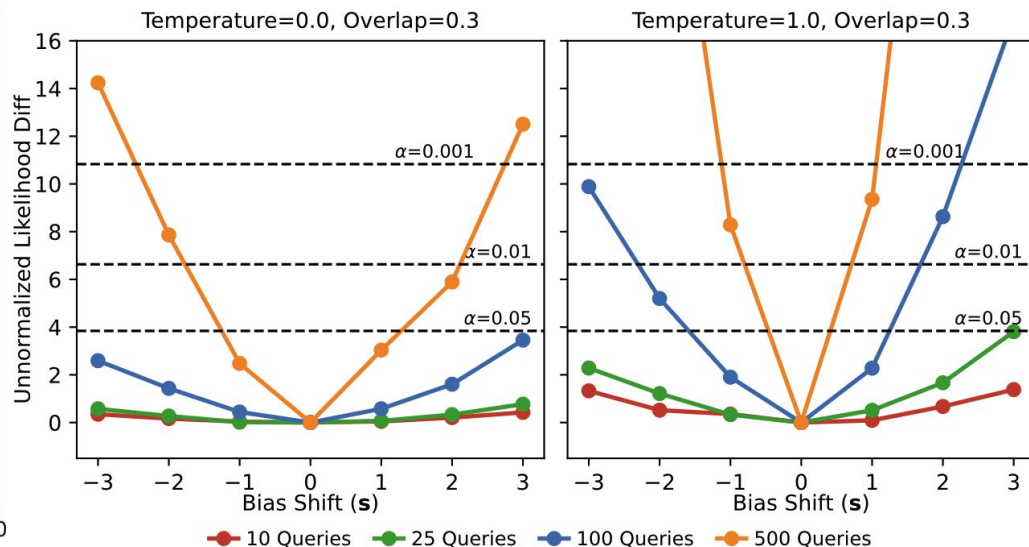
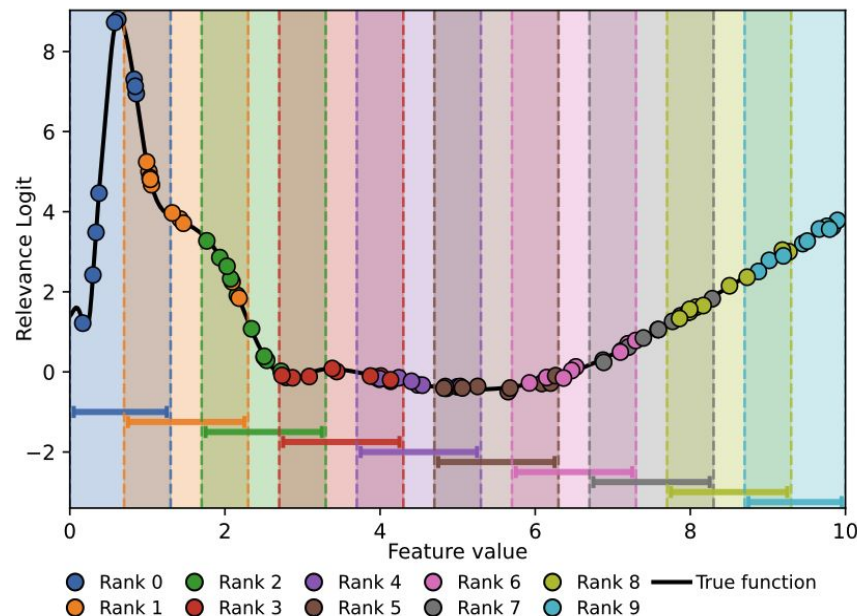


Effect of dataset parameters on Identifiability



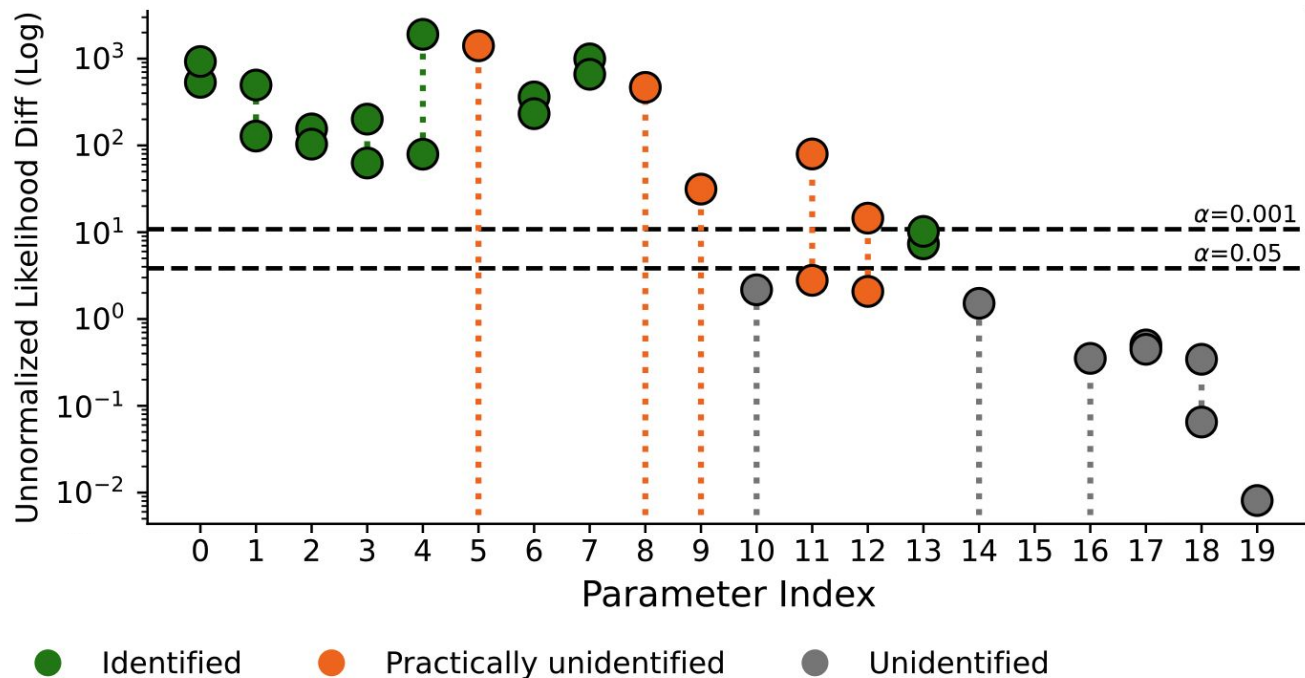
- Evaluating parameter identifiability when features do not overlap
- Randomization is required for reaching identifiability

Effect of dataset parameters on Identifiability



- Evaluating parameter identifiability when features do overlap
- We can reach identifiability, if we have enough samples

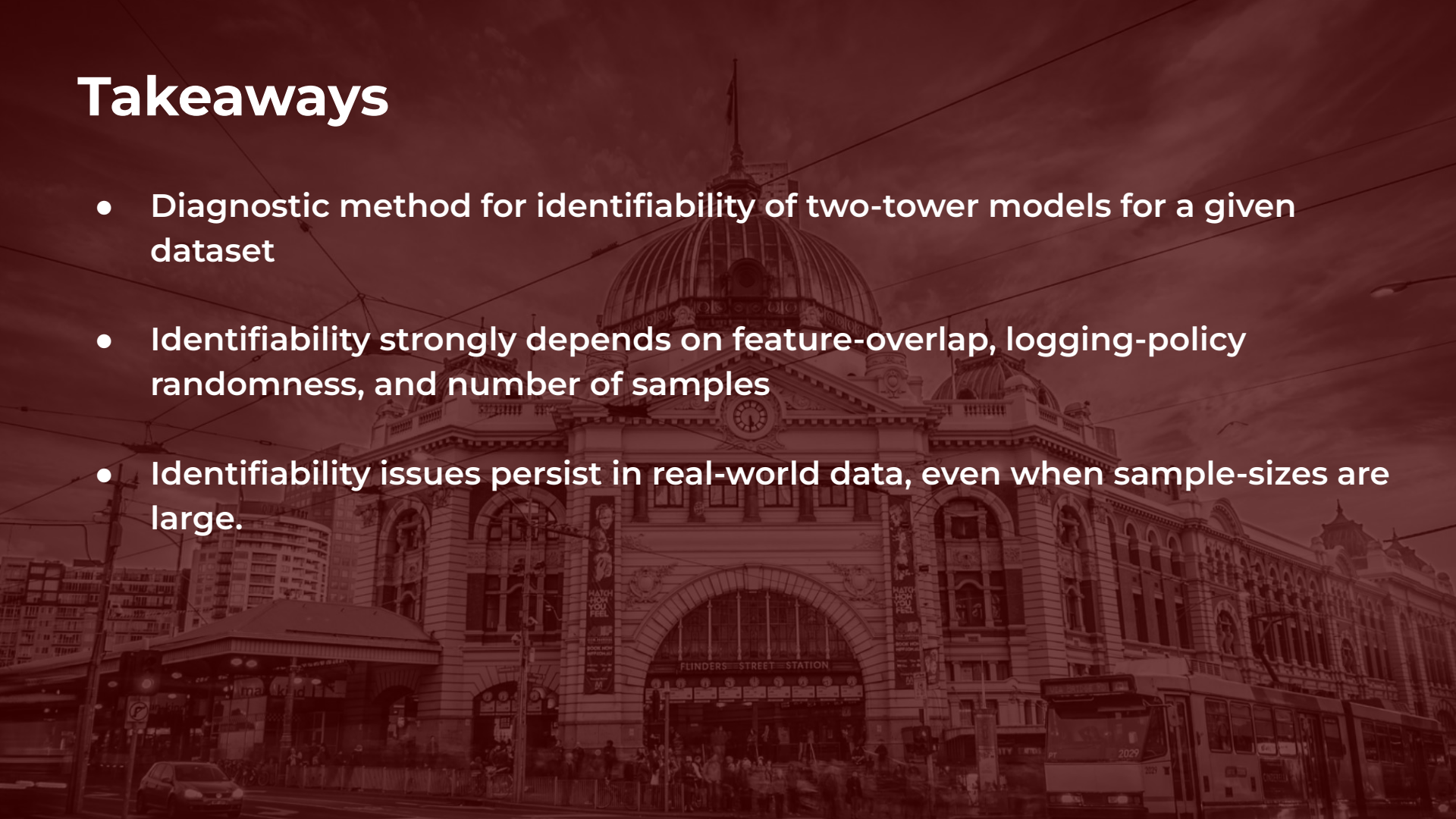
Diagnosing Identifiability in Real-World Data



- Several (low-rank) parameters are structurally unidentified
- Identifiability issues persist even in large click datasets.

Takeaways

- Diagnostic method for identifiability of two-tower models for a given dataset
- Identifiability strongly depends on feature-overlap, logging-policy randomness, and number of samples
- Identifiability issues persist in real-world data, even when sample-sizes are large.



Q&A

Read our Paper



Implementation



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